Concept approach

2.4 DC Electricity

This topic covers the definitions of various electrical quantities, for example, current and resistance, Ohm's law and non-ohmic materials, potential dividers, emf and internal resistance of cells, and negative temperature coefficient thermistors.

This topic may be studied using applications that relate to, for example, technology in space.

Students will be assessed on their ability to:	Suggested experiments
50 describe electric current as the rate of flow of charged particles and use the expression $I = \Delta Q / \Delta t$	
51 use the expression $V = W/Q$	
52 recognise, investigate and use the relationships between current, voltage and resistance, for series and parallel circuits, and know that these relationships are a consequence of the conservation of charge and energy	Measure current and voltage in series and parallel circuits Use ohmmeter to measure total resistance of series/parallel circuits
53 investigate and use the expressions $P = VI$, $W = VIt$. Recognise and use related expressions eg $P = I^2R$ and $P = V^2/R$	Measure the efficiency of an electric motor (see also outcome 17)
54 use the fact that resistance is defined by $R = V/I$ and that Ohm's law is a special case when $I \propto V$	
55 demonstrate an understanding of how ICT may be used to obtain current-potential difference graphs, including non-ohmic materials and compare this with traditional techniques in terms of reliability and validity of data	
56 interpret current-potential difference graphs, including non-ohmic materials	Investigate I-V graphs for filament lamp, diode and thermistor

Students will be assessed on their ability to:	Suggested experiments
57 investigate and use the relationship $R = \rho l/A$	Measure resisitivity of a metal and polythene
58 investigate and explain how the potential along a uniform current-carrying wire varies with the distance along it and how this variation can be made use of in a potential divider	Use a digital voltmeter to investigate 'output' of a potential divider
59 define and use the concepts of emf and internal resistance and distinguish between emf and terminal potential difference	Measure the emf and internal resistance of a cell eg a solar cell
60 investigate and recall that the resistance of metallic conductors increases with increasing temperature and that the resistance of negative temperature coefficient thermistors decreases with increasing temperature	Use of ohmmeter and temperature sensor
61 use $I = nqvA$ to explain the large range of resistivities of different materials	Demonstration of slow speed of ion movement during current flow
62 explain, qualitatively, how changes of resistance with temperature may be modelled in terms of lattice vibrations and number of conduction electrons	